fraction of the blood volume. The ratio of platelets to red blood cells in a healthy adult is about 1:10 to 1:20. Red blood cells are the most numerous blood cell, about 5,000,000 per microlitre. Red blood cells make up about 40% of our total blood volume, a measure called the hematocrit. White blood cells are the largest of the blood cells but also the fewest. There are normally only about 4,000 to 11,000 white blood cells per microlitre. Hence, statement of option (d) is correct, while other three statements are incorrect with reference to the blood in a normal person.

83. Which one of the following is synthesised in human body that dilates blood vessels and increases blood flow?

(a) Nitric oxide (b) Nitrous oxide
(c) Nitrogen dioxide (d) Nitrogen pentoxide

I.A.S. (Pre) 2024

Ans. (a)

Nitric oxide (NO) is synthesized in the human body, which is essential for many aspects of our health. The body produces nitric oxide from the amino acid L-arginine using the enzyme nitric oxide synthase (NOS). The primary site of NO synthesis is the inner layer of blood vessels (the endothelium), although other cell types also contribute to its production. Its main purpose is vasodilation, which entails relaxing the blood artery's inner muscles so that they can open up and improve circulation. Nitric oxide dilates blood vessels, raising blood supply and lowering blood pressure. Because it allows for correct and efficient blood, nutrient, and oxygen flow to all parts of the body, nitric oxide synthesis is essential for overall health. It also serves as a neurotransmitter and is involved in various functions, including neuronal activity and learning.

84. Consider the following :

Assertion (A) : Alum is applied to stop bleeding from cuts.

Reason (R) : Blood is a colloidal system containing colloidal particles. The Aluminium ions of Alum have high coagulating power so the blood gets coagulated.

Select the correct answer using the code given below:

Code :

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not a correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.P.C.S. (Mains) 2010

Ans. (a)

Alum is a specific crystalline substance and the aluminium ion present in it has high coagulating property. So, the alum is considered very good for the coagulation of blood when someone gets wounded and blood starts coming out from the wound.

85. Scientists of which country have developed an artificial blood that is a kind of plastic blood which could be given to any patient regardless of his blood group ?

(a) U.S.A. (b) Britain
(c) China (d) Norway

U.P.P.C.S. (Mains) 2006

Ans. (b)

Scientists from the University of Sheffield, Britain are developing an artificial 'plastic blood', which could act as a substitute for real blood in emergency situations. The 'plastic blood' have a huge impact on military applications in war zones. Because the artificial blood is made from a plastic, it is light to carry, does not need to be kept cool, can be kept for longer and easy to store. Doctors could store the substitute as a thick paste in a blood bag and then dissolve it in water just before giving it to patients – meaning it's easier to transport than blood.

VI. Excretory System

Notes

- **Excretion** is a process by which metabolic waste is eliminated from an organism.
- In vertebrates, this is primarily carried out by the lungs, kidneys and skin.
- This is in contrast with secretion, where the substance may have specific tasks after leaving the cell.
- Excretion is an essential process in all forms of life.

Classification of animals on the type of nitrogenous waste :

- The animals have been classified into three groups on the basis of the type of nitrogenous waste :

(a) Ammonotelic animals : Their main nitrogenous waste is **ammonia** e.g. certain protozoans, polychaete annelids, crustacean arthropods, molluscs (aplysia, sepia and octopus), fresh-water teleost fishes, tadpoles of amphibians and crocodiles.

(b) Ureotelic animals : These animal's main nitrogenous waste is **urea** e.g. elasmobranch fishes, amphibian and mammals.

(c) Uricotelic animals : Their main nitrogenous waste is **uric acid** e.g. insects, some gastropods, lizards, snakes and birds.

Excretion of nitrogenous waste in human :

- The nitrogenous waste in human is excreted by kidneys.
- The kidneys are bean-shaped organs which are present on each side of the vertebral column in the abdominal cavity.
- Humans have two kidneys and each is supplied with blood from the **renal artery**.
- The kidneys remove the nitrogenous waste from the blood such as urea as well as salts and excess water, and excrete them in the form of urine.
- This is done with the help of about one million **nephrons** present in the each kidney. Nephrons are the structural and functional unit of the kidney.
- The filtered blood is carried away from the kidney by the **renal vein**.
- Each kidney is attached to a **ureter**, a tube that carries excreted urine to the **urinary bladder**.
- The urinary bladder collects and stores the urine until urination.
- The urine collected in the bladder is passed into the external environment from the body through an opening called the **urethra**.
- The kidney's primary function is the elimination of waste from the bloodstream by the production of urine. They perform several homeostatic functions such as–
  (1) maintain volume of extracellular fluid.
  (2) maintain ionic balance in extracellular fluid.
  (3) maintain pH and osmolality of extracellular fluid.
  (4) regulate blood pressure in long term through maintenance of the extracellular fluid compartment.
  (5) excrete toxic metabolic by-products such as urea, ammonia and uric acid.
- The way the kidneys do this is with **nephrons**. These nephrons act as filters inside the kidneys.
- The kidney filter contains the needed materials and waste. The needed materials go back into the bloodstream and unneeded materials are excreted.
- In some cases, excess waste crystalizes as **kidney stones (calcium oxalate)**. They grow and become painful that may require surgery or lithotripsy treatments. Some stones are small enough to be forced into the urethra.
- Urea is formed in the liver but it is separated from the blood in the kidney by **ultrafiltration**.
- The urea is transported to kidney by circulatory system where it is filtered.
- The normal range for 24-hour urine volume is 800 to 2000 ml. per day with an average of about 1.5 litre per day (with a normal fluid take of about 2 litres per day).
- The urine is acidic with pH 6.0.

Dialyzer or Artificial Kidney

- A **dialyzer** is an artificial filter containing fine fibres. The fibres are hollow with microscopic pores in the wall, also known as a semipermeable membrane.
- Its function is to remove the excess waste and fluid from the blood when the patient's kidneys can no longer perform that task.
- The process of purifying blood in body is known as **dialysis**.

- Its pale yellow colour is due to the presence of pigment **urochrome** (urobilin).
- In urine, water about 91-96%; urea- about 2.6%; ions of unnecessary salts-about 2%; cretanin-about 0.3% and very small amount of uric acid and other waste materials in trace amount is found.
- The kidneys secrete a variety of hormones, including **erythropoietin**, **calcitriol** and **renin**.
- Most alcohol is broken down or metabolized by an enzyme by liver cells known as **alcohol dehydrogenase**. It breaks down alcohol into acetaldehyde and then another enzyme **aldehyde dehydrogenase**, rapidly breaks down acetaldehyde into acetate.
- From drugs and alcohol to unknown foreign substances, the **liver** helps filter and detoxify the materials not meant to be in our body. Ensuring toxins are safely removed from blood is one of the liver's most critical jobs.
- Sweat glands in the **skin** secrete a fluid waste called sweat, its primary functions are temperature control and pheromones release. Therefore, its role as a part of the excretory system is minimal. Sweating also maintains the level of salt in the body.
- Skin is considered an excretory organ because it excretes water, salt and urea from the body through sweat.

Question Bank

1. Which one of the following human organs is responsible for detoxification of alcohol ?

(a) Liver (b) Lung
(c) Heart (d) Kidney

U.P.P.C.S. (Mains) 2010

Ans. (a)

The liver is responsible for the detoxification of alcohol. Liver detoxify harmful substances through the complex chemical reactions.

2. A healthy human being excretes the following litres of urine in 24 hours –

(a) 1.5 (b) 3.0
(c) 6.0 (d) 9.0

U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (a)

Excretion of average urine in the adult human body is around 1.5 litres per day. About 91-96% of urine consist of water.

3. The yellow colour of human urine is due to a pigment called –

(a) Cytochrome (b) Urochrome
(c) Haemochrome (d) Phenolichrome

U.P. Lower Sub. (Pre) 2008

Ans. (b)

A pigment urochrome is the reason for the yellow colour of human urine. It is also known as urobilin.

4. With reference to the work of human kidney, consider the following statements –

1. After the waste is removed in the kidney, the cleaner blood is sent back through renal artery.

2. From Bowman's capsule, the filtered liquid passes through tiny tubes where much of the glucose is reabsorbed and sent back to the blood in the renal vein.

Which of these statements is/are correct?

(a) only 1 (b) only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2002

Ans. (b)

The renal artery carries blood that contains waste products to the nephrons for filtering. After waste products are removed, cleaner blood leaves the kidney by way of the renal vein. Thus, statement 1 is wrong while statement 2 is correct. From Bowman's capsule, the filtered liquid passes through tiny tubes where much of the glucose is reabsorbed and sent back to the blood in the renal vein.

5. The process by which blood is purified in human body is called :

(a) Dialysis (b) Haemolysis
(c) Osmosis (d) Paralysis

44th B.P.S.C. (Pre) 2000

Ans. (a)

Dialysis is a process for removing waste and excess water from the blood and is used primarily as an artificial replacement for lost kidney function in people with kidney failure. Dialysis filters out unwanted substances and fluids from the blood.

6. 'Dialysis' is related to ?

(a) Liver (b) Kidney
(c) Eyes (d) Brain

M.P.P.C.S. (Pre) 2004

Ans. (b)

See the explanation of above question.

7. The dialyzer is used for the work of –

(a) Heart (b) Kidney
(c) Liver (d) Lungs

U.P. Lower Sub. (Pre) 2002

Ans. (b)

The treatment options for kidney failure are peritoneal dialysis (PD), hemodialysis and kidney transplant. A dialyzer is an artificial kidney designed to provide controllable transfer of solutes and water across a semipermeable membrane separating flowing blood and dialysate streams.

8. Which of the following is correct ?

(a) All the blood in the body is absorbed through the kidneys.
(b) All the blood in the body passes through the kidneys.
(c) All the blood in the body is filtered through the kidneys
(d) All the blood in the body is made through the kidneys.

U.P.P.C.S. (Pre) 1994

Ans. (c)

Kidneys remove excess organic molecules from the blood and it is by this action that their best-known function is performed–the removal of waste products of metabolism. They serve the body as a natural filter of the blood and remove water-soluble wastes, such as urea and ammonium and they are also responsible for the reabsorption of water, glucose and amino acids.

9. Where is urea separated from the blood?

(a) Intestine (b) Stomach
(c) Spleen (d) Kidney

U.P. Lower Sub. (Pre) 2002

Ans. (d)

The process, ultrafiltration occurs at the barrier between the blood and the filtrate in the renal capsule in the kidney. The kidneys remove urea and other toxic wastes from the blood, forming a dilute solution called urine in the process.

10. The kidneys in human beings are a part of system for

(a) Nutrition (b) Transportation
(c) Excretion (d) Respiration

M.P.P.C.S. (Pre) 2017

Ans. (c)

In humans, kidneys are the main organ of the excretion system. Filtering the blood and removing unnecessary and waste products from the body is the basic function of kidney.

11. The average blood flow through kidneys per minute is

(a) 1000 cc (b) 1200 cc

(c) 200 cc (d) 500 cc

56th to 59th B.P.S.C. (Pre) 2015

Ans. (b)

Kidneys are the most important organ in our body. Kidneys help in purification of blood and also removes toxic materials from our body through urine. Our kidneys purify around 1500 litre of blood and excrete approximately 1.5 litre urine per day. About 1200 ml (cc) of blood flows through both the kidneys per minute and out of it about 1 ml of urine is formed per minute.

12. When kidneys fail to function, there is accumulation of–

(a) Fats in the body

(b) Proteins in the body

(c) Sugar in the blood

(d) Nitrogenous waste products in the blood

Uttarakhand P.C.S. (Pre) 2007

Ans. (d)

Metabolism refers to all of the body's chemical process, the digestion of food and the elimination of waste. The main nitrogenous wastes are ammonia, urea and uric acid. Urea is formed from gluconeogenesis of amino acids. Urea is one of the primary components of urine. When kidneys fail to function, there is accumulation of nitrogenous waste products in the blood.

13. Which of the following is not the normal function of the human kidney?

(a) Regulation of water level in the blood

(b) Regulation of sugar level in the blood

(c) Filter out urea

(d) Secretion of several hormones

U.P.P.C.S. (Pre) 2011

Ans. (b)

The main function of the pancreas is to produce insulin hormones. The pancreas plays an important role in digestion and in regulating blood sugar level in the blood, while other three options are the normal functions of human kidneys.

14. The major chemical compound found in human kidney stones is :

(a) Urea (b) Calcium carbonate

(c) Calcium oxalate (d) Calcium sulphate

I.A.S. (Pre) 1998

Ans. (c)

Calcium oxalate is a chemical compound that forms envelope-shaped crystals, known in plants as raphides. A major constituent of human kidney stones is calcium oxalate.

15. Kidney stones are formed due to :

(a) Precipitation of Proteins

(b) Crystalization of Oxalates

(c) Deposition of sand particles

(d) Deposition of Fat

U.P. R.O./A.R.O. (Pre) 2023

Ans. (b)

A kidney stone is a hard object that is made from chemicals in the urine. There are four types of kidney stones : calcium oxalate, uric acid, struvite, and crystine. The most common type of kidney stone which is created when calcium combines with oxalate in the urine. Oxalate is one type of substance that can form crystals in the urine. This can happen if there is too much oxalate, too little liquid, and the oxalate 'sticks' to calcium while urine is being made by the kidneys.

16. The 'stones' formed in human kidney consist mostly of :

(a) Calcium oxalate (b) Sodium acetate

(c) Magnesium sulphate (d) Calcium

I.A.S. (Pre) 2000

Ans. (a)

See the explanation of above question.

17. What amongst the following is responsible for the formation of stone in the human kidney?

(a) Calcium acetate (b) Calcium oxalate

(c) Sodium acetate (d) Sodium benzoate

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (b)

See the explanation of above question.

18. The element excreted through human sweat is :

(a) Sulfur (b) Iron

(c) Magnesium (d) Zinc

(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (e)

Sweat is produced by the skin in the form of liquid to regulate the body temperature. It is the part of the excretory function of the skin. Sodium, potassium, calcium, magnesium and many other trace elements (eg. zinc, copper, iron, chromium, nickel and lead) are excreted through human sweat.